

EXERCISE 10A

Solution 1: Let the fourth angle be x .

$$75^\circ + 90^\circ + 75^\circ + x = 360^\circ \quad [\text{sum of all the interior angles is } 360^\circ]$$

$$\Rightarrow 240^\circ + x = 360^\circ$$

$$\Rightarrow x = 360^\circ - 240^\circ$$

$$\Rightarrow x = 120^\circ$$

$\therefore$ Measure of the fourth angle is 120° .

Solution 2: Let the angles of the quadrilateral are $2x$, $4x$, $5x$ and $7x$

Since the sum of the angles of a quadrilateral is 360°

$$\therefore 2x + 4x + 5x + 7x = 360^\circ$$

$$\Rightarrow 18x = 360^\circ$$

$$\Rightarrow x = 20^\circ$$

So, the angles are

$$2x = 2 \times 20^\circ = 40^\circ$$

$$4x = 4 \times 20^\circ = 80^\circ$$

$$5x = 5 \times 20^\circ = 100^\circ$$

$$7x = 7 \times 20^\circ = 140^\circ$$

Solution 3: Since, $AB \parallel DC$. $\angle A$ and $\angle D$ are the interior angles on the same side of transversal line AD ,

$$\angle A + \angle D = 180^\circ$$

$$\Rightarrow \angle D = 180^\circ - \angle A$$

$$\Rightarrow \angle D = 180^\circ - 55^\circ$$

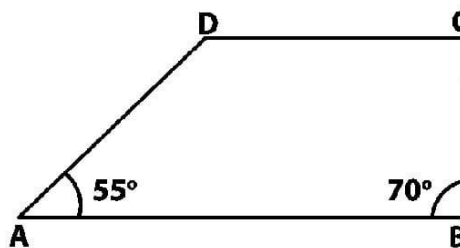
$$\Rightarrow \angle D = 125^\circ$$

Also,

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

$$\Rightarrow 55^\circ + 70^\circ + \angle C + 125^\circ = 360^\circ$$

$$\Rightarrow \angle C = 360^\circ - 250^\circ$$



$$\Rightarrow \angle C = 110^\circ$$

$\therefore \angle C$ and $\angle D$ are 110° and 125°

Solution 4:

(i) In $\triangle ADE$ and $\triangle BCE$,

$$AD = BC \quad [\text{Sides of a square}]$$

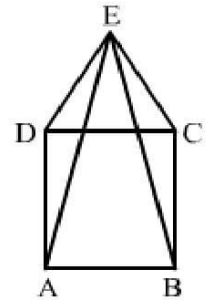
$$DE = EC \quad [\text{Sides of an equilateral triangle}]$$

$$\angle ADE = \angle BCE \quad [\angle ADE = \angle BCE = \angle ADE = \angle EDC + \angle CDA = 150^\circ]$$

By SAS congruence criterion, we get

$$\therefore \triangle ADE \cong \triangle BCE$$

$$\text{i.e., } AE = BE$$



(ii) In $\triangle EDA$,

$$\therefore ED = DA$$

$$\therefore \angle DEA = \angle DAE \quad \dots(1)$$

Also,

$$\angle EDA = \angle EDC + \angle CDA = 90^\circ + 60^\circ = 150^\circ$$

Now,

$$\angle EDA + \angle DAE + \angle DEA = 180^\circ$$

$$\Rightarrow 150^\circ + \angle DAE + \angle DEA = 180^\circ \quad [\text{From (1)}]$$

$$\Rightarrow 150^\circ + \angle DAE + \angle DAE = 180^\circ$$

$$\Rightarrow 2\angle DAE = 180^\circ - 150^\circ$$

$$\Rightarrow 2\angle DAE = 30^\circ$$

$$\Rightarrow \angle DAE = 15^\circ$$

Solution 5:

It is given that $BM \perp AC$ and $DN \perp AC$

From the figure, we have

$$\therefore \angle DON = \angle MOB \quad [\text{Vertically opposite angles}]$$

$$\angle DNO = \angle BMO \quad [\text{DN and MB are perpendiculars, so each of } 90^\circ]$$

$$BM = DN$$

[Given]

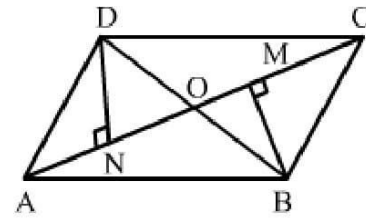
By AAS congruence criterion, we get

$$\triangle DNO \cong \triangle BMO$$

$$OD = OB$$

[CPCT]

Hence, AC bisects BD.



Solution 6:

(i) In $\triangle ABC$ and $\triangle ADC$,

$$AB = AD$$

[Given]

$$BC = DC$$

[Given]

$$AC = AC$$

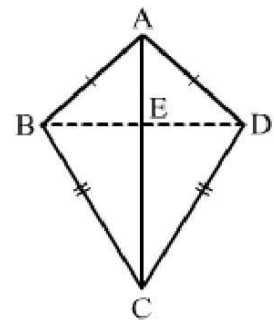
[Common]

By SSS congruence criterion, we get

$$\triangle ABC \cong \triangle ADC$$

$$\therefore \angle BAC = \angle DAC \text{ and } \angle BCA = \angle DCA \quad [\text{CPCT}]$$

Thus, AC bisects $\angle A$ and $\angle C$.



(ii)

Now, in $\triangle ABE$ and $\triangle ADE$, we have:

$$AB = AD$$

[Given]

$$\angle BAE = \angle DAE$$

$$AE = AE$$

[Common]

By SAS congruence criterion

$$\triangle ABE \cong \triangle ADE$$

$$\therefore BE = DE \quad [\text{CPCT}]$$

(iii) We have, $\triangle ABC \cong \triangle ADC$

$$\therefore \angle ABC = \angle ADC \quad [\text{CPCT}]$$

Solution 7:

(i)

From figure we can write

$$QB = BC - QC$$

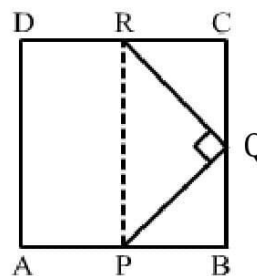
Since, ABCD is a square

$$\therefore BC = DC \text{ and } QC = DR$$

$$QB = CD - DR$$

$$\Rightarrow QB = DR + RC - DR$$

$$\Rightarrow QB = RC$$



(ii) In ΔRCQ and ΔQBP ,

$$PB = QC$$

[Given]

$$BQ = RC$$

$$\angle RCQ = \angle QBP$$

[90° each]

By SAS congruence criterion

$$\Delta RCQ \cong \Delta QBP$$

$$\therefore QR = PQ$$

[CPCT]

(iii) We know that,

$$\Delta RCQ \cong \Delta QBP \text{ and } QR = PQ$$

$\therefore$ In ΔRPQ ,

$$\angle QPR = \angle QRP = \frac{1}{2}(180^\circ - 90^\circ) = 45^\circ$$

Solution 8:

Join O with A, B, C, and D.

In ΔAOC ,

$$OA + OC > AC \quad \dots(1)$$

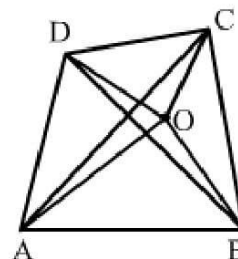
In ΔBOD ,

$$OB + OD > BD \quad \dots(2)$$

Adding (1) and (2), we get

$$(OA + OC) + (OB + OD) > (AC + BD)$$

$$\Rightarrow OA + OB + OC + OD > AC + BD$$



Solution 9:

(i) In $\triangle ABC$,

$$AB + BC > AC \quad \dots(1)$$

In $\triangle ACD$,

$$CD + DA > AC \quad \dots(2)$$

Adding (1) and (2), we get:

$$AB + BC + CD + DA > 2AC$$

(ii) In $\triangle ABC$,

$$AB + BC > AC$$

$$\Rightarrow AB + BC + CD > AC + CD \quad \dots(1)$$

In $\triangle ACD$

$$AC + CD > DA \quad \dots(2)$$

By substituting (2) in (1) we get

$$AB + BC + CD > DA$$

(iii) In $\triangle ABC$,

$$AB + BC > AC \quad \dots(1)$$

In $\triangle ACD$,

$$CD + DA > AC \quad \dots(2)$$

In $\triangle BCD$,

$$BC + CD > BD \quad \dots(3)$$

In $\triangle ABD$,

$$DA + AB > BD \quad \dots(4)$$

Adding (1), (2), (3) and (4)

$$2(AB + BC + CD + DA) > 2(AC + BD)$$

$$\Rightarrow (AB + BC + CD + DA) > (AC + BD)$$

Solution 10:

In $\triangle ABC$

$$\angle CAB + \angle B + \angle BCA = 180^\circ \quad \dots(1)$$

In $\triangle ACD$

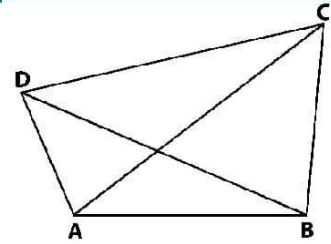
$$\angle DAC + \angle ACD + \angle D = 180^\circ \quad \dots (2)$$

Adding equations (1) and (2) we get

$$\angle CAB + \angle B + \angle BCA + \angle DAC + \angle ACD + \angle D = 180^\circ + 180^\circ$$

$$\Rightarrow \angle CAB + \angle DAC + \angle B + \angle BCA + \angle ACD + \angle D = 360^\circ$$

$$\Rightarrow \angle A + \angle B + \angle C + \angle D = 360^\circ$$



EXERCISE 10B

Solution 1: We know that in a parallelogram opposite angles are equal and sum of adjacent angles is 180° .

$$\therefore \angle A = \angle C = 72^\circ \text{ and } \angle B = \angle D$$

$$\angle A + \angle B = 180^\circ$$

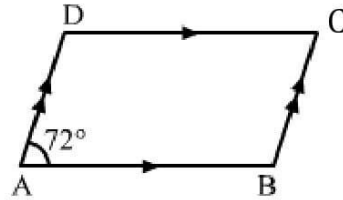
$$\Rightarrow \angle B = 180^\circ - \angle A$$

$$\Rightarrow \angle B = 180^\circ - 72^\circ$$

$$\Rightarrow \angle B = 108^\circ$$

$$\therefore \angle B = \angle D = 108^\circ$$

Hence, $\angle B = \angle D = 108^\circ$ and $\angle C = 72^\circ$



Solution 2:

We know that opposite angles are equal in parallelogram

$$\angle C = \angle A = 80^\circ$$

In parallelogram ABCD, $AD \parallel BC$

$$\therefore \angle DBC = \angle ADB = 60^\circ \quad [\text{Alternate interior angles}]$$

As $\angle DAB$ and $\angle ADC$ are adjacent angles,

$$\angle DAB + \angle ADC = 180^\circ$$

$$\Rightarrow \angle ADC = 180^\circ - \angle DAB$$

$$\Rightarrow \angle ADC = 180^\circ - 80^\circ$$

$$\Rightarrow \angle ADC = 100^\circ$$

$$\text{Also, } \angle ADC = \angle ADB + \angle CDB$$

$$\therefore \angle ADC = 100^\circ$$

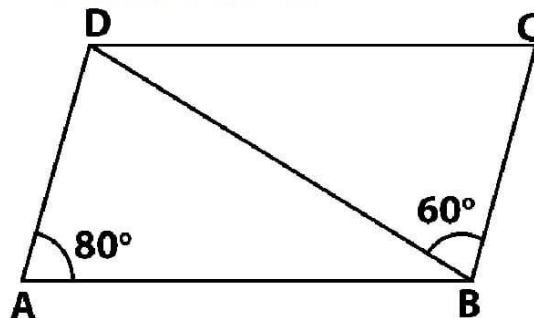
$$\Rightarrow \angle ADB + \angle CDB = 100^\circ$$

$$\Rightarrow 60^\circ + \angle CDB = 100^\circ \quad [\because \angle ADB = 60^\circ]$$

$$\Rightarrow \angle CDB = 100^\circ - 60^\circ$$

$$\Rightarrow \angle CDB = 40^\circ$$

Hence, $\angle CDB = 40^\circ$ and $\angle ADB = 60^\circ$



Solution 3:

It is given that ABCD is a parallelogram

So, $AD \parallel BC$, $AB = DC$ and $AD = BC$

From the figure,

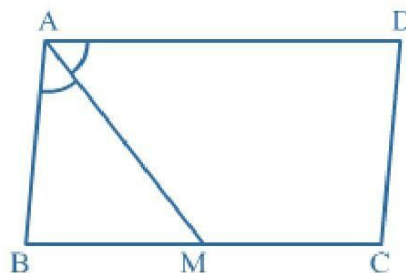
$$\angle DAM = \angle AMB$$

[$\because \angle DAM$ and $\angle AMB$ are alternate angles]

Now,

$$\angle BAM = \angle DAM$$

$$\Rightarrow \angle BAM = \angle AMB$$



Also, $BM = AB$

[sides opposite to equal angles]

And $AB = CD$

[opposite sides of a parallelogram]

$$\therefore BM = CD$$

$$\Rightarrow \frac{1}{2} BC = CD$$

[M is midpoint of BC]

$$\Rightarrow \frac{1}{2} AD = CD$$

[$BC = AD$]

$$AD = 2CD.$$

Therefore, it is proved that $AD = 2CD$.

Solution 4:

Since, ABCD is a parallelogram.

$$\therefore \angle A = \angle C \text{ and } \angle B = \angle D$$

[Opposite angles are equal in parallelogram]

$$\text{And } \angle A + \angle B = 180^\circ$$

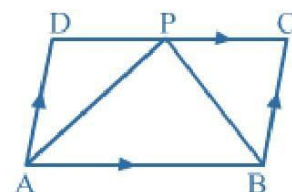
[Adjacent angles are supplementary]

$$\therefore \angle B = 180^\circ - \angle A$$

$$\Rightarrow \angle B = 180^\circ - 60^\circ = 120^\circ$$

[$\because \angle A = 60^\circ$]

$$\therefore \angle A = \angle C = 60^\circ \text{ and } \angle B = \angle D = 120^\circ$$



(i) In $\triangle APB$,

$$\angle PAB = \frac{60^\circ}{2} = 30^\circ \text{ [}\because \text{AP is bisector of } \angle A\text{]}$$

$$\angle PBA = \frac{120^\circ}{2} = 60^\circ \quad [\because PB \text{ is bisector of } \angle B]$$

$$\therefore \angle APB = 180^\circ - (30^\circ + 60^\circ) = 90^\circ$$

(ii) In $\triangle ADP$,

$$\angle PAD = 30^\circ \text{ and } \angle ADP = 120^\circ$$

$$\therefore \angle APD = 180^\circ - (30^\circ + 120^\circ) = 30^\circ$$

$$\text{Thus, } \angle PAD = \angle APD = 30^\circ$$

We know that the sides of an isosceles triangle are equal

$$DP = AD$$

$$\angle CPB = 60^\circ \text{ and } \angle C = 60^\circ$$

Now, by using angle sum property

$$\angle C^\circ + \angle CPB + \angle PBC = 180^\circ$$

$$\Rightarrow 60^\circ + 60^\circ + \angle PBC = 180^\circ$$

$$\Rightarrow \angle PBC = 180^\circ - (60^\circ + 60^\circ)$$

$$\Rightarrow \angle PBC = 180^\circ - 120^\circ$$

$$\Rightarrow \angle PBC = 60^\circ$$

We know that all the sides of an equilateral triangle are equal

$$PB = PC = BC$$

Therefore, it is proved that $AD = DP$ and $PB = PC = BC$.

(iii) $DC = DP + PC$

$$\Rightarrow DC = AD + BC \quad [AD = BC]$$

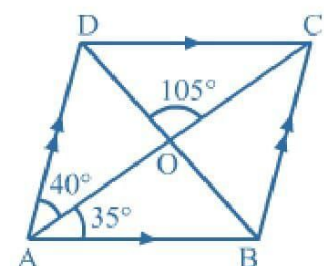
$$\Rightarrow DC = AD + AD$$

$$\Rightarrow DC = 2 AD$$

Solution 5:

ABCD is a parallelogram.

$$\therefore AB \parallel DC \text{ and } BC \parallel AD$$



(i) In $\triangle AOB$, $\angle BAO = 35^\circ$, $\angle AOB = \angle COD = 105^\circ$ [Vertically opposite angles]

Now, $\angle OAB + \angle AOB + \angle ABO = 180^\circ$

$$\Rightarrow \angle ABO = 180^\circ - (35^\circ + 105^\circ)$$

$$\Rightarrow \angle ABO = 40^\circ$$

(ii)

Since, $AB \parallel DC$ and BD is a transversal.

$\therefore \angle ODC$ and $\angle ABO$ are alternate interior angles.

$$\therefore \angle ODC = \angle ABO = 40^\circ$$

(iii) Since, $AB \parallel CD$ and AC is a transversal.

$$\angle ACB = \angle CAD = 40^\circ \quad [\text{Alternate interior angles}]$$

(iv) We know that

$$\angle CBD = \angle ABC - \angle ABD \quad \dots(1)$$

$$\text{Also, } \angle ABC = 180^\circ - \angle BAD \quad [\text{Adjacent angles are supplementary}]$$

$$\Rightarrow \angle ABC = 180^\circ - 75^\circ$$

$$\Rightarrow \angle ABC = 105^\circ$$

$$\Rightarrow \angle CBD = 105^\circ - \angle ABD \quad [\angle ABD = \angle ABO]$$

$$\Rightarrow \angle CBD = 105^\circ - 40^\circ$$

$$\Rightarrow \angle CBD = 65^\circ$$

Solution 6: We know that the opposite angles are equal in a parallelogram

$$\therefore \angle A = \angle C \text{ and } \angle B = \angle D \quad [\text{Opposite angles}]$$

$$\text{Also, } \angle A + \angle B = 180^\circ \quad [\text{Adjacent angles are supplementary}]$$

$$\therefore (2x + 25)^\circ + (3x - 5)^\circ = 180^\circ$$

$$\Rightarrow 5x^\circ + 20^\circ = 180^\circ$$

$$\Rightarrow 5x^\circ = 160^\circ$$

$$\Rightarrow x = 32$$

$$\therefore \angle A = (2 \times 32 + 25)^\circ = 89^\circ$$

$$\angle B = (3 \times 32 - 5)^\circ = 91^\circ$$

$$\angle C = 89^\circ$$

$$\angle D = 91^\circ$$

Solution 7: Let ABCD be a parallelogram.

$$\text{If } \angle A = x^\circ$$

Since, $\angle B$ is adjacent to A so, $\angle B$ can be written as $\frac{4}{5}x^\circ$

Now, $\angle A = \angle C$ and $\angle B = \angle D$ [Opposite angles]

$$\therefore \angle A = \angle C = x^\circ \text{ and } \angle B = \angle D = \left(\frac{4}{5}\right)x^\circ$$

$$\Rightarrow \angle A + \angle B + \angle C + \angle D = 360^\circ$$

$$\Rightarrow x^\circ + \left(\frac{4}{5}\right)x^\circ + x^\circ + \left(\frac{4}{5}\right)x^\circ = 360^\circ$$

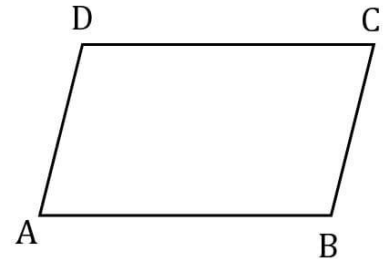
$$\Rightarrow 2x^\circ + \left(\frac{8}{5}\right)x^\circ = 360^\circ$$

$$\Rightarrow \left(\frac{18}{5}\right)x^\circ = 360^\circ$$

$$\Rightarrow x^\circ = \frac{360^\circ \times 5}{18}$$

$$\Rightarrow x^\circ = 100^\circ$$

$$\therefore \angle A = \angle C = 100^\circ \text{ and } \angle B = \angle D = \left(\frac{4}{5}\right) \times 100^\circ = 80^\circ$$



Solution 8:

Let ABCD be a parallelogram and $\angle A$ be the smallest angle.

$$\therefore \angle A = \angle C \text{ and } \angle B = \angle D \quad [\text{Opposite angles}]$$

$$\therefore \angle B = 2\angle A - 30^\circ$$

$$\text{Now, } \angle A + \angle B = 180$$

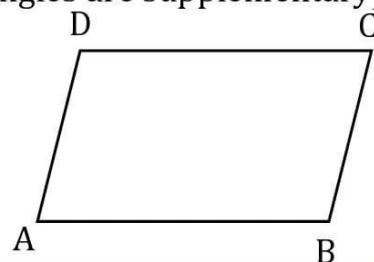
$$\Rightarrow \angle A + 2\angle A - 30^\circ = 180^\circ$$

$$\Rightarrow 3\angle A = 210^\circ$$

$$\Rightarrow x = 70^\circ$$

$$\therefore \angle B = 2 \times 70^\circ - 30^\circ = 110^\circ$$

[Adjacent angles are supplementary]



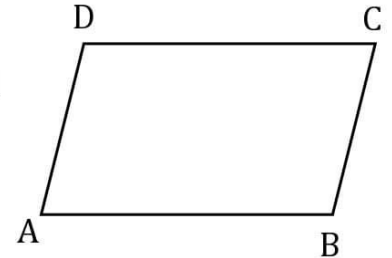
Hence, $\angle A = \angle C = 70^\circ$; $\angle B = \angle D = 110^\circ$

Solution 9:

Let, ABCD is a parallelogram.

We know that the opposite sides of a parallelogram are parallel and equal.

$\therefore AB = DC = 9.5 \text{ cm}$ and $BC = AD$



$\therefore$ Perimeter of ABCD = $AB + BC + CD + DA = 30 \text{ cm}$

$\Rightarrow 9.5 + BC + 9.5 + AD = 30$

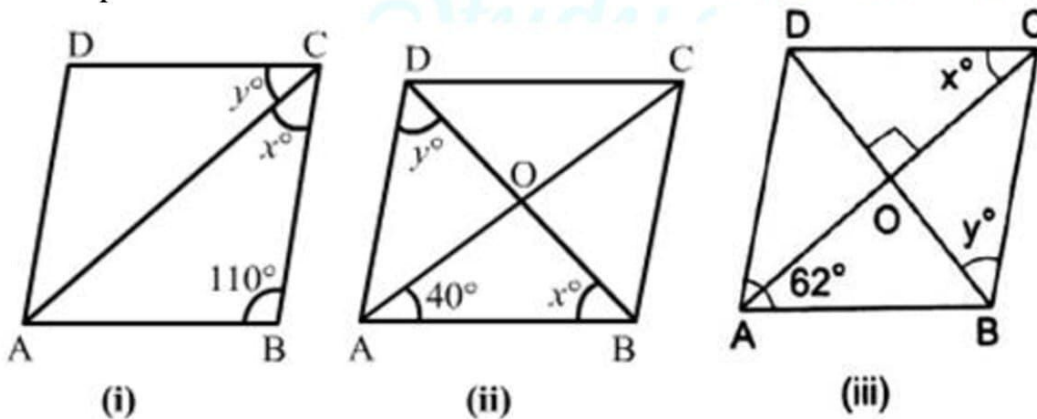
$\Rightarrow 19 + 2BC = 30$

$\Rightarrow 2BC = 11$

$\Rightarrow BC = 5.5 \text{ cm}$

Hence, $AB = DC = 9.5 \text{ cm}$ and $BC = DA = 5.5 \text{ cm}$

Solution 10: ABCD is a rhombus and a rhombus is also a parallelogram. A rhombus has four equal sides.



(i) In $\triangle ABC$,

$$\angle BAC = \angle BCA = \frac{1}{2}(180 - 110)^\circ = \frac{1}{2} \times 70^\circ = 35^\circ$$

$\therefore x = 35^\circ$

Now, $\angle B + \angle C = 180^\circ$

[Adjacent angles are supplementary]

$$\text{But } \angle C = x + y = 70^\circ$$

$$\Rightarrow y = 70^\circ - x$$

$$\Rightarrow y = 70^\circ - 35^\circ = 35^\circ$$

$$\therefore x = 35^\circ; y = 35^\circ$$

(ii) The diagonals of a rhombus are perpendicular bisectors of each other.

In $\triangle AOB$,

$$\angle OAB = 40^\circ,$$

$$\angle AOB = 90^\circ$$

$$\angle ABO = 180^\circ - (40^\circ + 90^\circ)$$

$$\Rightarrow \angle ABO = 180^\circ - 130^\circ$$

$$\Rightarrow \angle ABO = 50^\circ$$

$$\therefore x = 50^\circ$$

In $\triangle ABD$, $AB = AD$

$$\angle ABD = \angle ADB = 50^\circ$$

$$\therefore x = 50^\circ, y = 50^\circ$$

(iii) ABCD is a rhombus.

We know that rhombus is type of parallelogram whose all sides are equal.

So, in $\triangle DCB$,

$$DC = BC$$

$$\therefore \angle CDB = \angle CBD = y^\circ \quad [\text{base angles of isosceles triangle are equal}]$$

$$\text{Now, } x = \angle CAB \quad [\text{alternate angles with transversal AC}]$$

$$\therefore x = \frac{1}{2} \angle BAD$$

$$\Rightarrow x = \frac{1}{2} \times 62^\circ$$

$$\Rightarrow x = 31^\circ$$

In $\triangle DOC$,

We know sum of angles of triangle is 180°

$$\therefore \angle CDO + \angle DOC + \angle OCD = 180^\circ$$

$$\Rightarrow \angle CDO + 90^\circ + 31^\circ = 180^\circ$$

$$\Rightarrow \angle CDO = 59^\circ$$

$$\Rightarrow y = 59^\circ$$

Hence, $x = 31^\circ$ and $y = 59^\circ$

Solution 11: Let ABCD be a rhombus.

$$\therefore AB = BC = CD = DA$$

Here, AC and BD are the diagonals of ABCD,
where $AC = 24$ cm and $BD = 18$ cm.

Let the diagonals intersect each other at O.

We know that the diagonals of a rhombus are perpendicular bisectors of each other.

Consider $\triangle AOB$

$$\angle AOB = 90^\circ$$

$$AO = \frac{1}{2} AC = \frac{1}{2} \times 24 = 12 \text{ cm}$$

$$BO = \frac{1}{2} BD = \frac{1}{2} \times 18 = 9 \text{ cm}$$

Now,

$$AB^2 = OA^2 + OB^2 \quad [\text{Pythagoras theorem}]$$

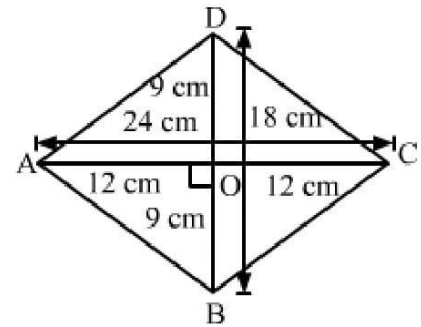
$$\Rightarrow AB^2 = (12)^2 + (9)^2$$

$$\Rightarrow AB^2 = 144 + 81 = 225$$

$$\Rightarrow AB = \sqrt{225}$$

$$\Rightarrow AB = 15 \text{ cm}$$

Therefore, the length of each side of the rhombus is 15 cm.



Solution 12: Let ABCD be a rhombus.

$$\therefore AB = BC = CD = DA = 10 \text{ cm}$$

Let AC and BD be the diagonals of ABCD.

Also, $AC = 16$ cm and O be the intersection point of the diagonals.

We know that the diagonals of a rhombus are perpendicular bisectors of each other.

$$AO = OC = \frac{1}{2} AC$$

$$\Rightarrow AO = OC = \frac{1}{2} \times 16 = 8 \text{ cm}$$

In $\triangle AOB$

$$AB^2 = AO^2 + OB^2$$

[By Pythagoras Theorem]

$$\Rightarrow 10^2 = 8^2 + OB^2$$

$$\Rightarrow OB^2 = 100 - 64$$

$$\Rightarrow OB^2 = 36$$

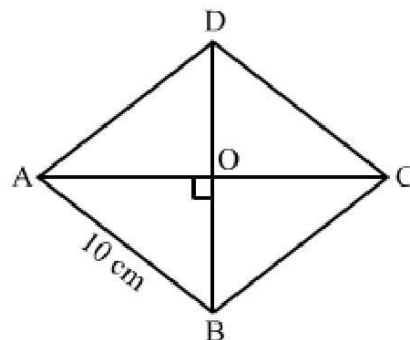
$$\Rightarrow OB = \sqrt{36}$$

$$OB = 6 \text{ cm}$$

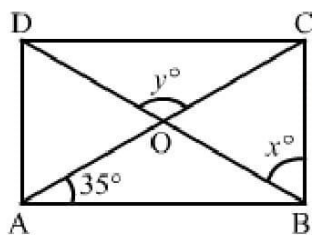
$$\therefore BD = 2 \times OB = 2 \times 6 = 12 \text{ cm}$$

Hence, the other diagonal of the rhombus is 12 cm.

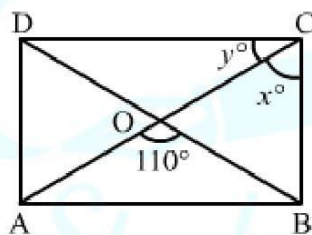
$$\therefore \text{Area of the rhombus} = \frac{1}{2} \times (12 \times 16) = 96 \text{ cm}^2$$



Solution 13:



(i)



(ii)

(i) Since, ABCD is a rectangle.

Therefore, in $\triangle AOB$

$$OA = OB$$

$$\therefore \angle OAB = \angle OBA = 35^\circ$$

$$\therefore x = 90^\circ - 35^\circ = 55^\circ$$

$$\text{And } \angle AOB = 180^\circ - (35^\circ + 35^\circ) = 110^\circ$$

$$\therefore y = \angle AOB = 110^\circ \quad [\text{Vertically opposite angles}]$$

$$\text{Hence, } x = 55^\circ \text{ and } y = 110^\circ$$

(ii) In $\triangle AOB$, we have:

$$OA = OB$$

Now,

$$\angle OAB = \angle OBA = \frac{1}{2} \times (180^\circ - 110^\circ) = 35^\circ$$

$$\therefore y = \angle BAC = 35^\circ \quad [\text{Interior alternate angles}]$$

$$\text{Also, } x^\circ = 90^\circ - y \quad [\because \angle C = 90^\circ = x^\circ + y^\circ]$$

$$\Rightarrow x = 90^\circ - 35^\circ = 55^\circ$$

Hence, $x = 55^\circ$ and $y = 35^\circ$

Solution 14:

Given: ABCD is a rhombus, DE is altitude which bisects AB i.e. $AE = EB$

In $\triangle AED$ and $\triangle BED$,

$$DE = DE \quad [\text{Common side}]$$

$$\angle DEA = \angle DEB = 90^\circ \quad [\text{Given}]$$

$$AE = EB \quad [\text{Given}]$$

$$\therefore \triangle AED \cong \triangle BED \quad [\text{By SAS congruence Criteria}]$$

$$\Rightarrow AD = BD \quad [\text{CPCT}]$$

$$\text{Also, } AD = AB \quad [\text{Sides of rhombus are equal}]$$

$$\Rightarrow AD = AB = BD$$

Thus, $\triangle ABD$ is an equilateral triangle.

$$\text{Therefore, } \angle A = 60^\circ$$

$$\Rightarrow \angle C = \angle A = 60^\circ \quad [\text{Opposite angles of rhombus are equal}]$$

$$\text{And, } \angle ABC + \angle BCD = 180^\circ \quad [\text{Adjacent angles of rhombus are supplementary}]$$

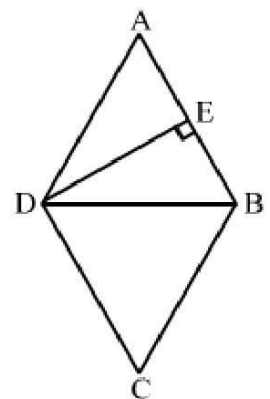
$$\Rightarrow \angle ABC + 60^\circ = 180^\circ$$

$$\Rightarrow \angle ABC = 180^\circ - 60^\circ$$

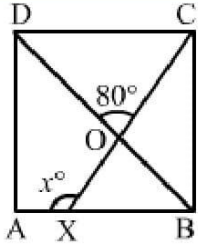
$$\Rightarrow \angle ABC = 120^\circ$$

$$\Rightarrow \angle ADC = \angle ABC = 120^\circ$$

Hence, the angles of rhombus are 60° , 120° , 60° and 120° .



Solution 15:



The angles of a square are bisected by the diagonals.

$$\therefore \angle OBX = 45^\circ$$

$$[\because \angle ABC = 90^\circ \text{ and } BD \text{ bisects } \angle ABC]$$

$$\text{And } \angle BOX = \angle COD = 80^\circ$$

[Vertically opposite angles]

$\therefore$ In $\triangle BOX$,

$$\angle AXO = \angle OBX + \angle BOX$$

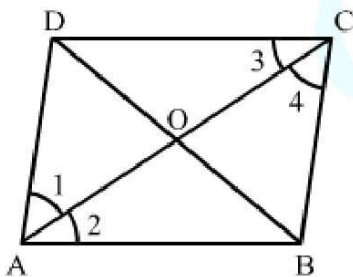
[Exterior angle of $\triangle BOX$]

$$\Rightarrow \angle AXO = 45^\circ + 80^\circ$$

$$\Rightarrow \angle AXO = 125^\circ$$

$$\therefore x = 125^\circ$$

Solution 16:



Given: A rhombus ABCD.

To prove: Diagonal AC bisects $\angle A$ as well as $\angle C$ and diagonal BD bisects $\angle B$ as well as $\angle D$.

Proof:

In $\triangle ABC$ and $\triangle ADC$

$$AB = AD \text{ and } BC = CD$$

[sides of rhombus are equal]

$$AC = AC$$

[Common]

By SSS congruence criterion

$$\triangle ABC \cong \triangle ADC$$

$$\therefore \angle BAC = \angle DAC \text{ and } \angle BCA = \angle DCA$$

[CPCT]

Therefore, AC bisects $\angle A$ as well as $\angle C$.

In $\triangle BAD$ and $\triangle BCD$

$AB = BC$ and $AD = CD$

$BD = BD$

By SSS congruence criterion

$\triangle BAD \cong \triangle BCD$

$\therefore \angle ABD = \angle CBD$ and $\angle ADB = \angle CDB$

Therefore, BD bisects $\angle B$ as well as $\angle D$.

Solution 17:

Given: In a parallelogram $ABCD$, $AM = CN$.

To prove: AC and MN bisect each other.

Construction: Join AN and MC .

Proof:

In parallelogram $ABCD$, $AB \parallel DC$.

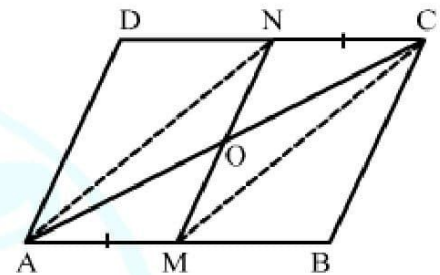
$\Rightarrow AM \parallel NC$

Also, $AM = CN$ [Given]

Thus, $AMCN$ is a parallelogram.

Since, diagonals of a parallelogram bisect each other.

Hence, AC and MN bisect each other.



Solution 18:

We have:

$\angle B = \angle D$ [Opposite angles of parallelogram $ABCD$]

$AD = BC$ and $AB = DC$ [Opposite sides of parallelogram $ABCD$]

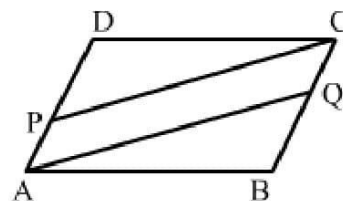
Also, $AD \parallel BC$ and $AB \parallel DC$

It is given that $AP = \frac{1}{3}AD$ and $CQ = \frac{1}{3}BC$

$\therefore AP = CQ$ [$\because AD = BC$]

Also,

$$DP = AD - PA = AD - \frac{1}{3}AD = \frac{2}{3}AD$$



$$BQ = BC - CQ = BC - \frac{1}{3} BC = \frac{2}{3} BC$$

We know that $AD = BC$

$$\therefore DP = BQ$$

In $\triangle DPC$ and $\triangle BQA$,

$$AB = CD,$$

$$\angle B = \angle D$$

$$DP = QB$$

By SAS congruence criterion

$$\triangle DPC \cong \triangle BQA$$

$$\therefore PC = QA$$

Thus, in quadrilateral AQCP,

$$AP = CQ \quad \dots(i)$$

$$PC = QA \quad \dots(ii)$$

$\therefore$ AQCP is a parallelogram.

Solution 19: In $\triangle ODF$ and $\triangle OBE$, we have:

$$OD = OB$$

[Diagonals bisect each other]

$$\angle DOF = \angle BOE$$

[Vertically opposite angles]

$$\angle FDO = \angle OBE$$

[Alternate interior angles]

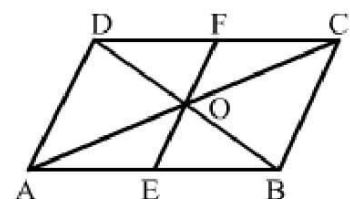
By ASA congruence criterion

$$\triangle ODF \cong \triangle OBE$$

$$\therefore OF = OE$$

[CPCT]

Hence, proved.



Solution 20: From the figure

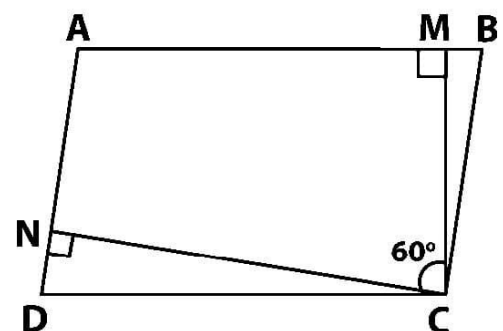
$$\angle DCM = \angle DCN + \angle MCN$$

$$\Rightarrow 90^\circ = \angle DCN + 60^\circ$$

$$\Rightarrow \angle DCN = 90^\circ - 60^\circ$$

$$\Rightarrow \angle DCN = 30^\circ$$

In $\triangle DCN$



$$\angle DNC + \angle DCN + \angle D = 180^\circ$$

By substituting values in the above equation

$$\Rightarrow 90^\circ + 30^\circ + \angle D = 180^\circ$$

On further calculation

$$\Rightarrow \angle D = 180^\circ - 90^\circ - 30^\circ$$

$$\Rightarrow \angle D = 180^\circ - 120^\circ$$

$$\Rightarrow \angle D = 60^\circ$$

We know that the opposite angles of parallelogram are equal

$$\angle B = \angle D = 60^\circ$$

It can be written as

$$\Rightarrow \angle A + \angle B = 180^\circ$$

$$\Rightarrow \angle A + 60^\circ = 180^\circ$$

$$\Rightarrow \angle A = 180 - 60^\circ$$

$$\Rightarrow \angle A = 120^\circ$$

$$\therefore \angle A = \angle C = 120^\circ$$

Therefore, $\angle B = \angle D = 60^\circ$ and $\angle A = \angle C = 120^\circ$.

Solution 21:

To prove:

- (i) ABCD is a square,
- (ii) diagonal BD bisects $\angle B$ as well as $\angle D$.

Proof:

- (i) We know that ABCD is a rectangle in which diagonal AC bisects $\angle A$ as well as $\angle C$.

$$\angle BAC = \angle DAC \quad \dots(1)$$

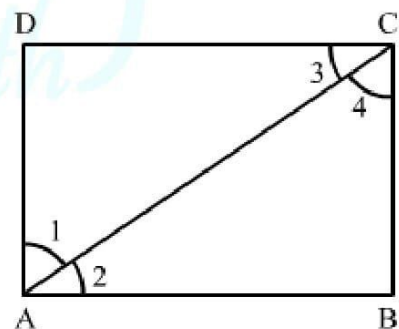
$$\angle BCA = \angle DCA \quad \dots(2)$$

We know that every rectangle is a parallelogram

$\therefore AB \parallel DC$ and AC is a transversal

$\therefore$ Also, $\angle BAC = \angle DCA$ [Alternate angles]

$$\Rightarrow \angle DAC = \angle DCA$$



In $\triangle ADC$

$AD = CD$ [Opposite sides of equal angles are equal]

Since ABCD is a rectangle

$\therefore AB = BC$ and $CD = AD$

$AB = BC = CD = AD$

Therefore, it is proved that ABCD is a square.

(ii) In $\triangle BAD$ and $\triangle BCD$

$AB = CD$ [Sides of a square]

$AD = BC$ [Sides of a square]

$BD = BD$ [Common]

By SSS congruence criterion

$\triangle BAD \cong \triangle BCD$

$\therefore \angle ABD = \angle CBD$

$\angle ADB = \angle CDB$ [CPCT]

Therefore, diagonal BD bisects $\angle B$ as well as $\angle D$.

Solution 22:

In $\triangle ODC$ and $\triangle OEB$,

$DC = BE$ [$\because DC = AB$ and $AB = BE$]

$\angle COD = \angle BOE$ [Vertically opposite angles]

$\angle OCD = \angle OBE$ [Alternate interior angles]

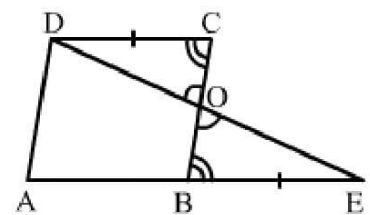
By AAS congruence criterion

$\triangle ODC \cong \triangle OEB$

$\Rightarrow OC = OB$ [CPCT]

We know that $BC = OC + OB$.

$\therefore ED$ bisects BC .



Solution 23: In parallelogram ABCD, $AB \parallel DC$

$\angle DCE = \angle EBF$ [Alternate interior angles]

In $\triangle DCE$ and $\triangle BFE$,

$$\angle DCE = \angle EBF$$

[Proved above]

$$\angle DEC = \angle BEF$$

[Vertically opposite angles]

Also, $BE = CE$

[Given]

By ASA congruence criterion

$$\therefore \triangle DCE \cong \triangle BFE$$

[By ASA congruence rule]

$$\therefore DC = BF$$

[CPCT]

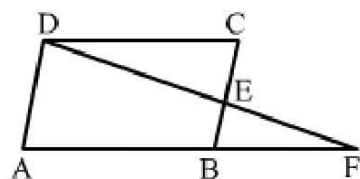
But $DC = AB$, as $ABCD$ is a parallelogram.

$$\therefore DC = AB = BF \quad \dots(i)$$

$$\text{Now, } AF = AB + BF \quad \dots(ii)$$

$$\Rightarrow AF = AB + AB = 2AB$$

Hence, proved.



Solution 24:

We know that $l \parallel m$ and t is a transversal.

From the figure, alternate angles

$$\Rightarrow \angle APR = \angle PRD$$

$$\Rightarrow \frac{1}{2} \angle APR = \frac{1}{2} \angle PRD$$

Since, PS and RQ are the bisectors of $\angle APR$ and $\angle PRD$

$$\angle SPR = \angle PRQ$$

Hence, PR intersects PS and RQ at points P and R respectively

$$\therefore PS \parallel RQ$$

Similarly, $SR \parallel PQ$

Therefore, $PQRS$ is a parallelogram.

Now,

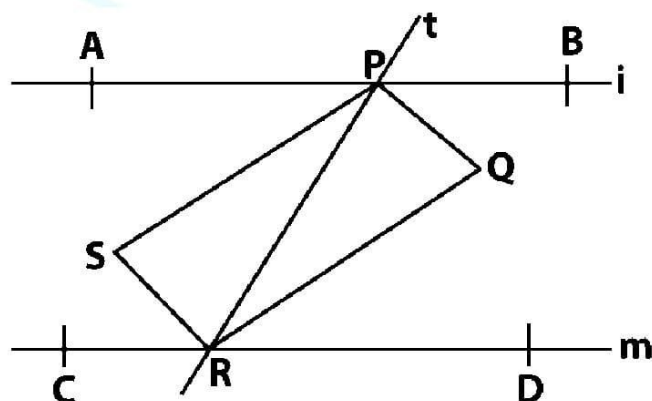
$$\angle BPR + \angle PRD = 180^\circ$$

[Interior angles are supplementary]

$$\Rightarrow 2\angle QPR + 2\angle QRP = 180^\circ$$

[PQ and RQ are the bisectors of $\angle BPR$ and $\angle PRD$]

$$\Rightarrow \angle QPR + \angle QRP = 90^\circ \quad \dots(1)$$



In $\triangle PQR$,

$$\angle PQR + \angle QPR + \angle QRP = 180^\circ$$

$$\Rightarrow \angle PQR + 90^\circ = 180^\circ$$

$$\Rightarrow \angle PQR = 180^\circ - 90^\circ$$

$$\Rightarrow \angle PQR = 90^\circ$$

Since, PQRS is a parallelogram

$$\angle PQR = \angle PSR = 90^\circ$$

$$\angle SPQ + \angle PQR = 180^\circ \quad [\text{adjacent angles are supplementary}]$$

$$\Rightarrow \angle SPQ + 90^\circ = 180^\circ$$

$$\Rightarrow \angle SPQ = 180^\circ - 90^\circ$$

$$\Rightarrow \angle SPQ = 90^\circ$$

Therefore, it is proved that the quadrilateral formed by the bisectors of interior angles is a rectangle.

Solution 25:

In square ABCD, $AK = BL = CM = DN$.

To prove: KLMN is a square.

Proof:

It is given that $AK = BL = CM = DN$

Since, ABCD is a square

$$\therefore BK = CL = DM = AN \dots\dots (1)$$

In $\triangle AKN$ and $\triangle BLK$

$$AK = BL \quad [\text{Given}]$$

$$\angle A = \angle B = 90^\circ \quad [\because \text{ABCD is Square}]$$

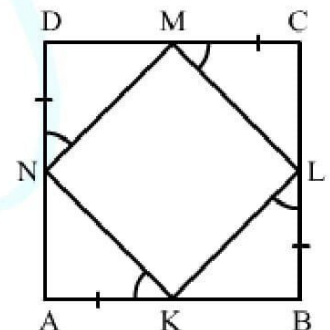
$$AN = BK \quad [\text{From equation (1)}]$$

By SAS congruence criterion

$$\triangle AKN \cong \triangle BLK$$

$$\therefore \angle AKN = \angle BLK \text{ and } \angle ANK = \angle BKL \quad [\text{CPCT}]$$

Now,



$$\angle AKN + \angle ANK = 90^\circ \quad \dots(2)$$

$$\angle BLK + \angle BKL = 90^\circ \quad \dots(3)$$

Adding equation (2) and (3),

$$\angle AKN + \angle ANK + \angle BLK + \angle BKL = 90^\circ + 90^\circ$$

$$\Rightarrow 2\angle ANK + 2\angle BLK = 180^\circ$$

$$\Rightarrow \angle ANK + \angle BLK = 90^\circ$$

$$\Rightarrow \angle NKL = 90^\circ$$

Similarly,

$$\angle KLM = \angle LMN = \angle MNK = 90^\circ$$

Therefore, it is proved that KLMN is a square.

Solution 26: From figure,

$BC \parallel QA$ and $CA \parallel QB$

BCQA is a parallelogram.

$$\therefore BC = QA \quad \dots(i)$$

Also, $BC \parallel AR$ and $AB \parallel CR$.

i.e., BCRA is a parallelogram.

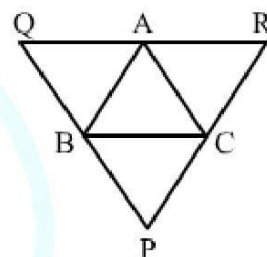
$$\therefore BC = AR \quad \dots(ii)$$

Now, $QR = QA + AR$

$$\Rightarrow QR = BC + BC \quad [\text{From (i) and (ii)}]$$

$$\Rightarrow QR = 2BC$$

$$\therefore BC = \frac{1}{2}QR$$



Solution 27: From the figure,

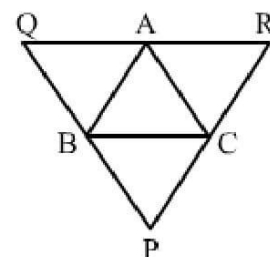
$BC \parallel QA$ and $CA \parallel QB$

$\therefore$ BCQA is a parallelogram.

$$BC = QA \quad \dots(i)$$

Similarly, $BC \parallel AR$ and $AB \parallel CR$

$\therefore$ BCRA is a parallelogram.



$$BC = AR \quad \dots(ii)$$

$$\text{Now, } QR = QA + AR$$

$$\Rightarrow QR = BC + BC \quad [\text{From equation (i) and (ii)}]$$

$$\Rightarrow QR = 2BC$$

$$\therefore BC = \frac{1}{2}QR$$

Similarly,

$$CA = \frac{1}{2}PQ \text{ and } AB = \frac{1}{2}PR$$

$$\text{Perimeter of } \triangle ABC = AB + BC + CA$$

$$= \frac{1}{2}QR + \frac{1}{2}PQ + \frac{1}{2}PR$$

$$= \frac{1}{2}(PR + QR + PQ) \quad \dots (iii)$$

$$\text{Perimeter of } \triangle PQR = PQ + QR + PR \quad \dots(iv)$$

From equation (iii) and (iv), we get

$$\text{Perimeter of } \triangle PQR = 2 \times \text{Perimeter of } \triangle ABC$$

EXERCISE 10C

Solution 1:

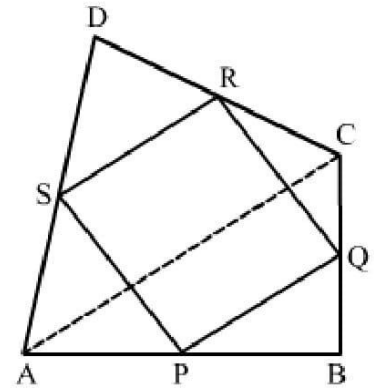
Given: In quadrilateral ABCD, P, Q, R and S
are respectively the midpoints of the sides AB, BC, CD and DA.

To prove:

(i) $PQ \parallel AC$ and $PQ = \frac{1}{2}AC$

(ii) $PQ \parallel SR$

(iii) PQRS is a parallelogram.



Proof:

(i) In $\triangle ABC$,

Since, P and Q are the mid points of sides AB and BC, respectively. By using mid-points theorem, we get

$$\Rightarrow AC \parallel PQ \text{ and } PQ = \frac{1}{2}AC \quad \dots (1)$$

(ii) In $\triangle ADC$,

Since, S and R are the mid-points of AD and DC, respectively

By using mid-points theorem, we get

$$\Rightarrow SR \parallel AC \text{ and } SR = \frac{1}{2}AC \quad \dots (2)$$

From equation (1) and (2), we get

$$PQ \parallel SR$$

(iii) From equation (1) and (2), we get

$$PQ = SR = \frac{1}{2}AC$$

Since, PQ and SR are parallel and equal.

Hence, PQRS is a parallelogram.

Solution 2:

Given: In an isosceles right $\triangle ABC$, $CMPN$ is a square.

To prove: P bisects the hypotenuse AB i.e., $AP = PB$.

Proof:

In $\triangle ABC$,

$AC = BC$ [$\because \triangle ABC$ is an isosceles right triangle, right angled at C]

In square $CMPN$,

$$\therefore CM = MP = PN = CN$$

Now,

$$AC = BC$$

$$\Rightarrow AN + NC = CM + MB$$

$$\Rightarrow AN = MB \quad \dots(1) \quad [\because CN = CM]$$

In $\triangle ANP$ and $\triangle PMB$,

$$AN = MB \quad [\text{From (1)}]$$

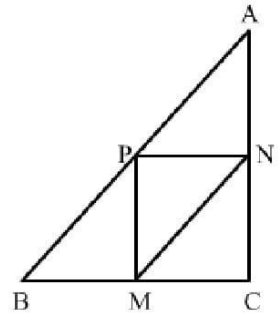
$$\angle ANP = \angle PMB = 90^\circ$$

$$PN = PM \quad [\text{Sides of square } CMPN]$$

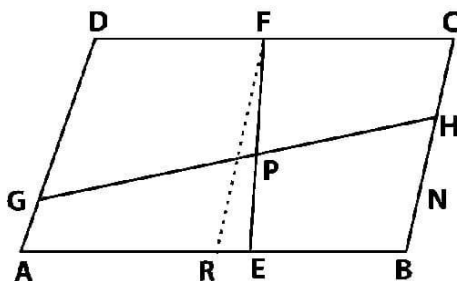
$\therefore$ By SAS congruence criteria,

$$\triangle ANP \cong \triangle BMP$$

$$\therefore AP = PB \quad [\text{By CPCT}]$$



Solution 3:



In parallelogram $ABCD$, know that AD , EF and BC are three line-segment and DC , AB are the two transversals. So, the intercepts made by the line on AB and CD are equal

$$\therefore AE = EB \text{ and } DF = FC$$

Now, we prove that FE is parallel to AD by using the method of contradiction

Assuming FE is not parallel to AD

Construct FR parallel to AD

From Intercept theorem, we get

$$AR = RB \text{ as } DF = FC,$$

$$\text{Also, } AE = EB$$

Since, AB cannot have two mid points i.e. R and E

$\therefore$ Our assumption is not correct

Hence, $AD \parallel EF \parallel BC$

By using the Intercept theorem, we know that GH is a transversal and the intercepts made by AD, EF and BC on GH are equal as $DF = FC$.

$$\therefore GP = PH. \text{ (Proved)}$$

Solution 4:

Given: A parallelogram ABCD

To prove: MN is bisected at O

Proof:

In $\triangle OAM$ and $\triangle OCN$,

$$OA = OC$$

[Diagonals of parallelogram bisect each other]

other]

$$\angle AOM = \angle CON$$

[Vertically opposite angles]

$$\angle MAO = \angle OCN$$

[Alternate interior angles]

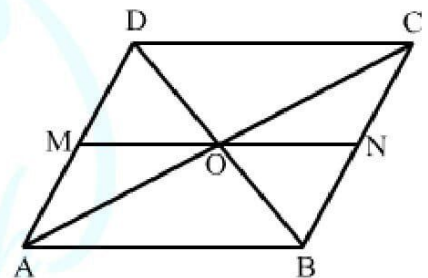
$\therefore$ By ASA congruence criterion,

$$\triangle OAM \cong \triangle OCN$$

$$\therefore OM = ON$$

[CPCT]

Hence, MN is bisected at O.



Solution 5:

Given: In trapezium PQRS, $PQ \parallel SR$, M is the midpoint of PS and $MN \parallel PQ$.

To prove: N is the midpoint of QR.

Construction: Join QS.

Proof:

In ΔSPQ ,

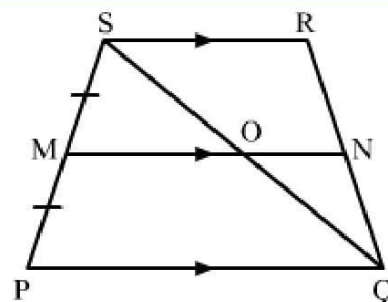
Since, M is the mid-point of SP and $MO \parallel PQ$.

$\therefore$ O is the mid-point of SQ. [By Mid-point theorem]

Similarly, in ΔSRQ ,

Since, O is the mid-point of SQ and $ON \parallel SR$ [SR $\parallel$ PQ and MN $\parallel$ PQ]

$\therefore$ N is the mid-point of QR. [By Mid-point theorem]



Solution 6:

Given: In parallelogram PQRS, PQ = 12 cm and PS = 9 cm.

The bisector of $\angle SPQ$ meets SR at M.

Let $\angle SPQ = 2x$.

$\Rightarrow \angle SRQ = 2x$ and $\angle TPQ = x$.

Also, $PQ \parallel SR$

$\Rightarrow \angle TMR = \angle TPQ = x$.

In ΔTMR , $\angle SRQ$ is an exterior angle.

$\Rightarrow \angle SRQ = \angle TMR + \angle MTR$

$\Rightarrow 2x = x + \angle MTR$

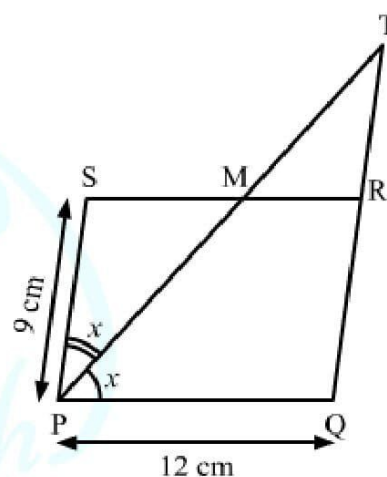
$\Rightarrow \angle MTR = x$

$\Rightarrow \Delta TPQ$ is an isosceles triangle.

$\Rightarrow TQ = PQ = 12$ cm

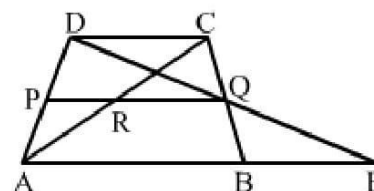
Now,

$RT = TQ - QR = TQ - PS = 12 - 9 = 3$ cm



Solution 7:

Given: $AB \parallel DC$, $AP = PD$ and $BQ = CQ$



(i) In $\triangle QCD$ and $\triangle QBE$, we have:

$\angle DQC = \angle BQE$	[Vertically opposite angles]
$\angle DCQ = \angle EBQ$	[Alternate angles, as $AE \parallel DC$]
$BQ = CQ$	[Q is the midpoints]
$\therefore \triangle QCD \cong \triangle QBE$	
Hence, $DQ = QE$	[CPCT]

(ii) Now, in $\triangle ADE$, P and Q are the midpoints of AD and DE, respectively.

$$\begin{aligned}\therefore PQ &\parallel AE, PQ \parallel AB \\ \Rightarrow PQ &\parallel AB \parallel DC \\ \Rightarrow AB &\parallel PR \parallel DC\end{aligned}$$

(iii) From the figure we know that the lines PR, AB and DC are cut by AC and AD respectively.

Based on the intercept theorem we know that $AR = RC$.

Therefore, it is proved that $AR = RC$.

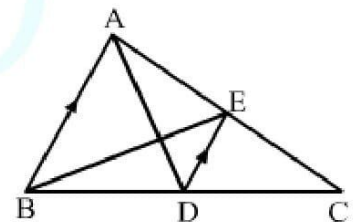
Solution 8: In $\triangle ABC$, $DE \parallel AB$ and D is the midpoint of BC

Based on the midpoint theorem we know that

E is the midpoint of AC

So we know that BE is the median of $\triangle ABC$ drawn through B.

Therefore, it is proved that BE is also a median of $\triangle ABC$.



Solution 9:

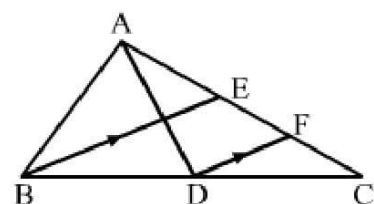
In $\triangle CBE$, D is the midpoint of BC and DF is parallel to BE.

By using the midpoint theorem, we can write

$$F \text{ is the midpoint of } EC, \text{ and } CF = \frac{1}{2} EC$$

Also, BE is the median through B

$$\therefore CF = \frac{1}{2} \times \left[\frac{1}{2} AC \right]$$



$$\Rightarrow CF = \frac{1}{4} AC$$

Solution 10: In $\triangle ABC$,

D, E and F are the midpoints of sides AB, BC and CA, respectively.

As, D and E are the mid points of sides AB, and BC of $\triangle ABC$.

$\therefore DE \parallel AC$ [By midpoint theorem]

Similarly, $DF \parallel BC$ and $EF \parallel AB$.

Therefore, ADEF, BDFE and DFCE are all parallelograms.

We know that a diagonal divides the parallelogram into two equal parts

Now, DE is the diagonal of the parallelogram BDFE.

$\therefore \triangle BDE \cong \triangle FED$

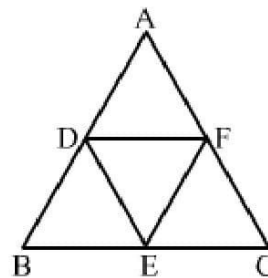
DF is the diagonal of the parallelogram ADEF.

$\therefore \triangle DAF \cong \triangle FED$

EF is the diagonal of the parallelogram DFCE.

$\therefore \triangle EFC \cong \triangle FED$

So, all the four triangles are congruent.



Solution 11:

Consider $\triangle AFE$ and $\triangle DFE$

Using the midpoint theorem

$$AF = \frac{1}{2} AB = ED$$

$$AE = \frac{1}{2} AC = FD$$

$$FE = FE \quad [\text{Common}]$$

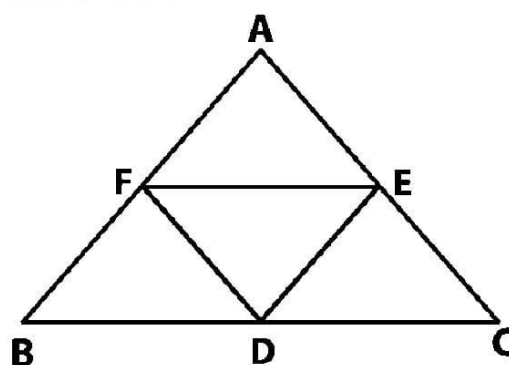
By SSS congruence criterion

$$\triangle AFE \cong \triangle DFE$$

$$\angle A = \angle FDE \quad [\text{CPCT}]$$

In the same way we get $\angle B = \angle DEF$ and $\angle C = \angle DFE$.

Therefore, it is proved that $\angle EDF = \angle A$, $\angle DEF = \angle B$ and $\angle DFE = \angle C$.



Solution 12:

Let ABCD be the rectangle and P, Q, R and S be the midpoints of AB, BC, CD and DA, respectively.

Join AC, a diagonal of the rectangle.

In $\triangle ABC$, P and Q are the mid points of AB and BC

By using the midpoint theorem

$$PQ \parallel AC \text{ and } PQ = \frac{1}{2} AC$$

Also, In $\triangle ADC$

$$RS \parallel AC \text{ and } RS = \frac{1}{2} AC$$

$\therefore$ we can write $PQ \parallel RS$ and

$$PR = RS \quad \left\{ \text{Each} = \frac{1}{2} AC \right\} \quad \dots (1)$$

So, PQRS is a parallelogram.

Now, in $\triangle SAP$ and $\triangle QBP$, we have:

$$AS = BQ$$

$$\angle A = \angle B = 90^\circ$$

$$AP = BP$$

$$\text{i.e., } \triangle SAP \cong \triangle QBP$$

$$\therefore PS = PQ \quad \dots (2)$$

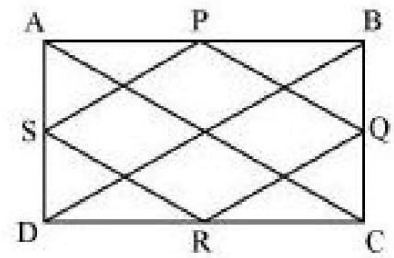
Similarly, $\triangle SDR \cong \triangle QCR$

$$\therefore RS = RQ \quad \dots (3)$$

From (1), (2) and (3), we have:

$$PQ = RS = SR = RQ$$

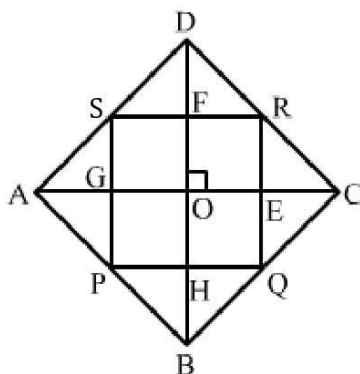
Hence, PQRS is a rhombus.



Therefore, it is proved that the quadrilateral formed by joining the midpoints of the pairs of adjacent sides of a rectangle is a rhombus.

Solution 13: Let ABCD be a rhombus and P, Q, R and S be the midpoints of AB, BC, CD and DA, respectively.

Join the diagonals, AC and BD.



In $\triangle ABC$,

$$PQ \parallel AC \text{ and } PQ = \frac{1}{2} AC \quad [\text{By midpoint theorem}]$$

Again, in $\triangle DAC$, the points S and R are the midpoints of AD and DC, respectively.

$$\therefore SR \parallel AC \text{ and } SR = \frac{1}{2} AC \quad [\text{By midpoint theorem}]$$

Now, $PQ \parallel AC$ and $SR \parallel AC \Rightarrow PQ \parallel SR$

$$\text{Also, } PQ = SR \quad [\text{Each equal to } \frac{1}{2} AC] \quad \dots (i)$$

So, PQRS is a parallelogram.

We know that the diagonals of a rhombus bisect each other at right angles.

$$\therefore \angle EOF = 90^\circ$$

Now, $RQ \parallel DB$

$$\Rightarrow RE \parallel FO$$

Also, $SR \parallel AC$

$$\Rightarrow FR \parallel OE$$

$\therefore$ OERF is a parallelogram.

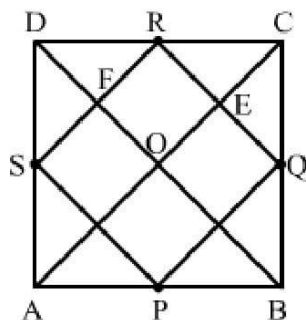
$$\text{So, } \angle FRE = \angle EOF = 90^\circ \quad [\text{Opposite angles are equal}]$$

Thus, PQRS is a parallelogram with $\angle R = 90^\circ$.

$\therefore$ PQRS is a rectangle.

Solution 14: Let ABCD be a square and P, Q, R and S be the midpoints of AB, BC, CD and DA, respectively.

Join the diagonals AC and BD. Let BD cut SR at F and AC cut RQ at E. Let O be the intersection point of AC and BD.



In $\triangle ABC$,

$$\therefore PQ \parallel AC \text{ and } PQ = \frac{1}{2} AC \quad [\text{By midpoint theorem}]$$

Again, in $\triangle DAC$, the points S and R are the midpoints of AD and DC, respectively.

$$\therefore SR \parallel AC \text{ and } SR = \frac{1}{2} AC \quad [\text{By midpoint theorem}]$$

Now, $PQ \parallel AC$ and $SR \parallel AC$

$$\Rightarrow PQ \parallel SR$$

$$\text{Also, } PQ = SR \quad [\text{Each equal to } \frac{1}{2} AC]$$

So, PQRS is a parallelogram.

Now, in $\triangle SAP$ and $\triangle QBP$,

$$AS = BQ$$

$$\angle A = \angle B = 90^\circ$$

$$AP = BP$$

$$\therefore \triangle SAP \cong \triangle QBP$$

$$PS = PQ \quad \dots(\text{ii})$$

Similarly, $\triangle SDR \cong \triangle RCQ$

$$\therefore SR = RQ \quad \dots(\text{iii})$$

From (i), (ii) and (iii)

$$PQ = PS = SR = RQ \quad \dots(\text{iv})$$

We know that the diagonals of a square bisect each other at right angles.

$$\therefore \angle EOF = 90^\circ$$

Now, $RQ \parallel DB$

$\Rightarrow RE \parallel FO$

Also, $SR \parallel AC$

$\Rightarrow FR \parallel OE$

$\therefore OERF$ is a parallelogram.

So, $\angle FRE = \angle EOF = 90^\circ$ [Opposite angles are equal]

Thus, PQRS is a parallelogram with $\angle R = 90^\circ$ and $PQ = PS = SR = RQ$.

$\therefore$ PQRS is a square.

Solution 15: Let ABCD be the quadrilateral in which P, Q, R, and S are the midpoints of sides AB, BC, CD, and DA, respectively.

Join PQ, QR, RS, SP and BD. BD is a diagonal of ABCD.

In $\triangle ABD$, S and P are the midpoints of AD and AB, respectively.

$\therefore SP \parallel BD$ and $SP = \frac{1}{2}BD \dots (i)$ [By midpoint theorem]

In $\triangle BCD$,

$QR \parallel BD$ and $QR = \frac{1}{2}BD \dots (ii)$ [By midpoint theorem]

From equations (i) and (ii), we get

$SP \parallel BD \parallel QR$

$\therefore SP \parallel QR$ and $SP = QR$ [Each equal to $\frac{1}{2}BD$]

In quadrilateral SPQR, one pair of the opposite sides is equal and parallel to each other.

$\therefore$ SPQR is a parallelogram.

We know that the diagonals of a parallelogram bisect each other.

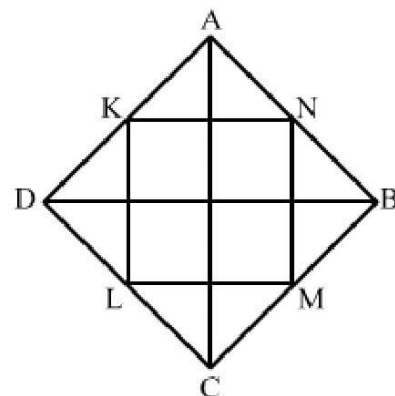
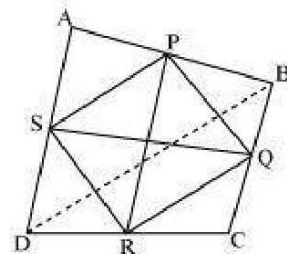
$\therefore$ PR and QS bisect each other.

Solution 16:

Given: In quadrilateral ABCD, $BD = AC$ and K, L, M and N are the mid-points of AD, CD, BC and AB, respectively.

To prove: KLMN is a rhombus.

Proof:



In $\triangle ADC$,

Since, K and L are the mid-points of sides AD and CD, respectively.

$$\therefore KL \parallel AC \text{ and } KL = \frac{1}{2}AC \quad \dots (1)$$

In $\triangle ABC$, M and N are the mid-points of sides BC and AB, respectively.

$$\therefore NM \parallel AC \text{ and } NM = \frac{1}{2}AC \quad \dots (2)$$

From (1) and (2), we get

$$KL = NM \text{ and } KL \parallel NM$$

But this a pair of opposite sides of the quadrilateral KLMN.

$\therefore$ KLMN is a parallelogram.

Now, in $\triangle ABD$, K and N are the mid-points of sides AD and AB, respectively.

$$\therefore KN \parallel BD \text{ and } KN = \frac{1}{2}BD \quad \dots (3)$$

$$\text{But } BD = AC \quad [\text{Given}]$$

$$\Rightarrow \frac{1}{2}BD = \frac{1}{2}AC$$

$$\Rightarrow KN = NM \quad [\text{From (2) and (3)}]$$

But these are a pair of adjacent sides of the parallelogram KLMN.

Hence, KLMN is a rhombus.

Solution 17:

Given: In quadrilateral ABCD, $AC \perp BD$. P, Q, R and S are the mid-points of AB, BC, CD and AD, respectively.

To prove: PQRS is a rectangle.

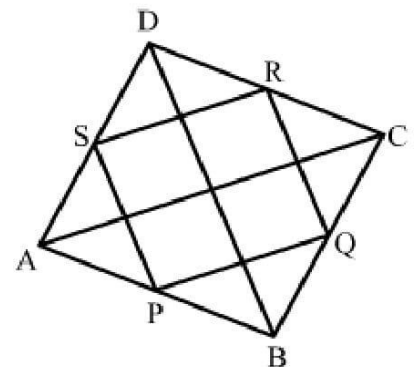
Proof:

In $\triangle ABC$, P and Q are mid-points of AB and BC, respectively.

$$\therefore PQ \parallel AC \text{ and } PQ = \frac{1}{2}AC \quad \dots (1) \quad [\text{midpoint theorem}]$$

In $\triangle ACD$, R and S are mid-points of sides CD and AD, respectively.

$$\therefore SR \parallel AC \text{ and } SR = \frac{1}{2}AC \quad \dots (2) \quad [\text{Midpoint theorem}]$$



From (1) and (2), we get

$$PQ \parallel SR \text{ and } PQ = SR$$

But this is a pair of opposite sides of the quadrilateral PQRS,

$\therefore$ PQRS is parallelogram.

In $\triangle BCD$, Q and R mid-points of BC and CD, respectively.

$$\therefore QR \parallel BD \text{ and } QR = \frac{1}{2}BD \dots (3) \quad [\text{Mid - point theorem}]$$

From (2) and (3), we get

$$SR \parallel AC \text{ and } QR \parallel BD$$

Now, $AC \perp BD$ [Given]

$$\therefore RS \perp QR$$

Hence, PQRS is a rectangle.

Solution 18:

Given: In quadrilateral ABCD, $AC = BD$ and $AC \perp BD$. P, Q, R and S are the mid-points of AB, BC, CD and AD, respectively.

To prove: PQRS is a square.

Construction: Join AC and BD.

Proof:

In $\triangle ABC$,

$\therefore$ P and Q are mid-points of AB and BC, respectively.

$$\therefore PQ \parallel AC \text{ and } PQ = \frac{1}{2}AC \dots (1) \quad [\text{Mid - point theorem}]$$

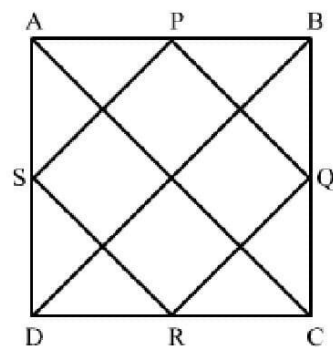
Similarly, in $\triangle ACD$,

$\therefore$ R and S are mid-points of sides CD and AD, respectively.

$$\therefore SR \parallel AC \text{ and } SR = \frac{1}{2}AC \dots (2) \quad [\text{Mid - point theorem}]$$

From (1) and (2), we get

$$PQ \parallel SR \text{ and } PQ = SR$$



But this a pair of opposite sides of the quadrilateral PQRS.

$\therefore$ PQRS is parallelogram.

Now, $\triangle BCD$, Q and R mid-points of sides BC and CD, respectively.

$$\therefore QR \parallel BD \text{ and } QR = \frac{1}{2}BD \quad \dots (3) \quad [\text{Mid - point theorem}]$$

From (2) and (3), we get

$$RS \parallel AC \text{ and } QR \parallel BD$$

$$\text{But, } AC \perp BD \quad [\text{Given}]$$

$$\therefore RS \perp QR$$

But this a pair of adjacent sides of the parallelogram PQRS.

$\therefore$ PQRS is a rectangle.

$$\text{Again, } AC = BD \quad [\text{Given}]$$

$$\Rightarrow \frac{1}{2}AC = \frac{1}{2}BD$$

From equation (2) and (3), we get

$$\Rightarrow SR = QR$$

But this a pair of adjacent sides of the rectangle PQRS.

Hence, PQRS is a square.

Multiple Choice Questions (MCQ)

Solution 1: (b) 73°

Let the measure of the fourth angle be x .

$$80^\circ + 95^\circ + 112^\circ + x = 360^\circ$$

$$\Rightarrow 287^\circ + x = 360^\circ$$

$$\Rightarrow x = 73^\circ$$

Hence, the measure of the fourth angle is 73° .

Solution 2: (b) 60°

Let the angles are $3x$, $4x$, $5x$ and $6x$.

Since the sum of the angles of a quadrilateral is 360° ,

$$3x + 4x + 5x + 6x = 360^\circ$$

$$\Rightarrow 18x = 360^\circ$$

$$\Rightarrow x = 20^\circ$$

The angles are 60° , 80° , 100° and 120°

Hence, the smallest angle is 60° .

Solution 3: (c) 45°

$$\angle B = 180^\circ - \angle A$$

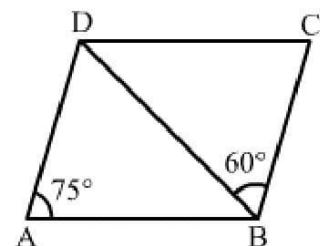
$$\Rightarrow \angle B = 180^\circ - 75^\circ = 105^\circ$$

$$\text{Now, } \angle B = \angle ABD + \angle CBD$$

$$\Rightarrow 105^\circ = \angle ABD + 60^\circ$$

$$\Rightarrow \angle ABD = 105^\circ - 60^\circ = 45^\circ$$

$$\Rightarrow \angle ABD = \angle BDC = 45^\circ \quad [\text{Alternate angles}]$$



Solution 4: (a) 40°

We know that diagonals of rhombus bisect each other at 90° .

Then, in $\triangle BOC$,

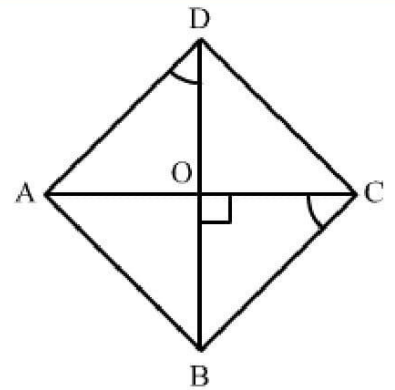
$$90^\circ + 50^\circ + \angle OBC = 180^\circ \quad [\text{Angle sum property of triangle}]$$

$$\Rightarrow \angle OBC = 180^\circ - 140^\circ$$

$$\Rightarrow \angle OBC = 40^\circ$$

$$\text{But } \angle OBC = \angle ADB \quad [\text{Alternate interior angles}]$$

$$\text{Thus, } \angle ADB = 40^\circ$$



Solution 5: (d) Rectangle.

Solution 6: (d) rhombus

Solution 7: (a) 10 cm

Let ABCD be the rhombus.

$$\therefore AB = BC = CD = DA$$

Here, AC and BD are the diagonals of ABCD, where AC = 16 cm and BD = 12 cm.

Let the diagonals intersect each other at O.

We know that the diagonals of a rhombus are perpendicular bisectors of each other.

$$\text{Now, } AB^2 = OA^2 + OB^2 \quad [\text{Pythagoras theorem}]$$

$$\Rightarrow AB^2 = (8)^2 + (6)^2$$

$$\Rightarrow AB^2 = 64 + 36 = 100$$

$$\Rightarrow AB = 10 \text{ cm}$$

Hence, the side of the rhombus is 10 cm.

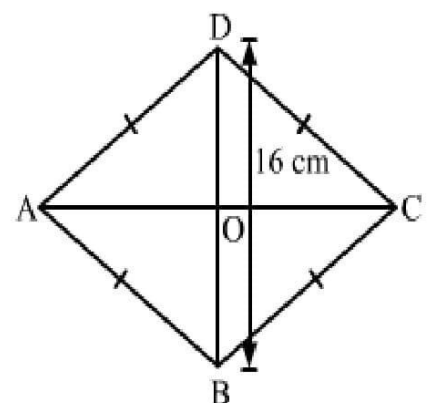
Solution 8: (b) 12 cm

Explanation:

Let ABCD be the rhombus.

$$\therefore AB = BC = CD = DA = 10 \text{ cm}$$

Let AC and BD be the diagonals of the rhombus.



Let AC be x and BD be 16 cm and O be the intersection point of the diagonals.

We know that the diagonals of a rhombus are perpendicular bisectors of each other.

$\therefore \triangle AOB$ is a right-angled triangle in which

$$OA = \frac{1}{2} AC = \frac{1}{2} x \text{ and } OB = \frac{1}{2} BD = \frac{1}{2} \times 16 = 8 \text{ cm.}$$

$$\text{Now, } AB^2 = OA^2 + OB^2 \quad [\text{Pythagoras theorem}]$$

$$\Rightarrow 10^2 = OA^2 + 8^2$$

$$\Rightarrow OA^2 = 36$$

$$\Rightarrow OA^2 = 6^2$$

$$\Rightarrow OA = 6$$

$$\therefore x = 2 \times 6 = 12 \text{ cm}$$

Solution 9:

In ABCD, $\angle OAD = 35^\circ$.

$$\Rightarrow \angle OAB = 90^\circ - 35^\circ = 55^\circ \quad [\because \angle BAD = 90^\circ]$$

In $\triangle OAB$,

$$OA = OB \quad [\text{Diagonals of a rectangle are equal and bisect each other}]$$

$$\Rightarrow \angle OAB = \angle OBA = 55^\circ \quad [\text{Angles opposite to equal sides are equal}]$$

Now, in $\triangle ODA$,

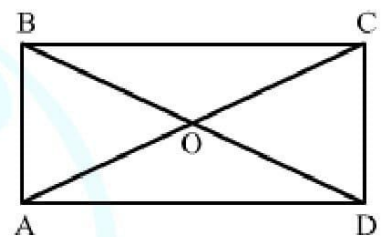
$$55^\circ + 55^\circ + \angle DOA = 180^\circ \quad [\text{Angle sum property of a triangle}]$$

$$\Rightarrow \angle DOA = 180^\circ - 110^\circ$$

$$\Rightarrow \angle DOA = 70^\circ$$

Thus, the acute angle between the diagonals is 70° .

Hence, the correct option is (b).



Solution 10: (c) Rectangle

$$\angle A = \angle B$$

$$\text{Then } \angle A + \angle B = 180^\circ$$

$$\Rightarrow 2\angle A = 180^\circ$$

$$\Rightarrow \angle A = 90^\circ$$

$$\Rightarrow \angle A = \angle B = \angle C = \angle D = 90^\circ$$

$\therefore$ The parallelogram is a rectangle.

Solution 11: (b) 50°

$$\angle C = 70^\circ \text{ and } \angle D = 30^\circ$$

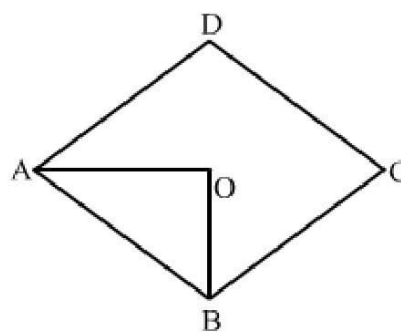
$$\text{Then } \angle A + \angle B = 360^\circ - (70 + 30)^\circ = 260^\circ$$

$$\therefore \frac{1}{2}(\angle A + \angle B) = \frac{1}{2} \times 260^\circ = 130^\circ$$

In ΔAOB ,

$$\angle AOB = 180^\circ - \left[\frac{1}{2}(\angle A + \angle B) \right]$$

$$\Rightarrow \angle AOB = 180^\circ - 130^\circ = 50^\circ$$



Solution 12: (d) 90°

Sum of two adjacent angles = 180°

$$\text{Now, sum of angle bisectors of two adjacent angles} = \frac{1}{2} \times 180^\circ = 90^\circ$$

$$\therefore \text{Intersection angle of bisectors of two adjacent angles} = 180^\circ - 90^\circ = 90^\circ$$

Solution 13: (c) Rectangle

Solution 14: (d) quadrilateral whose opposite angles are supplementary

In quadrilateral ABCD, AS, BQ, CQ and DS are angle bisectors of angles A, B, C and D, respectively.

$$\angle QPS = \angle APB \quad \dots(1) \quad [\text{Vertically opposite angles}]$$

In $\triangle APB$,

$$\angle APB + \angle PAB + \angle ABP = 180^\circ \quad [\text{Angle sum property of triangle}]$$

$$\Rightarrow \angle APB + \frac{1}{2}\angle A + \frac{1}{2}\angle B = 180^\circ$$

$$\Rightarrow \angle APB = 180^\circ - \frac{1}{2}(\angle A + \angle B) \quad \dots (2)$$

From (1) and (2), we get

$$\angle QPS = 180^\circ - \frac{1}{2}(\angle A + \angle B) \quad \dots (3)$$

Similarly,

$$\angle QRS = 180^\circ - \frac{1}{2}(\angle C + \angle D) \quad \dots (4)$$

From (3) and (4), we get

$$\begin{aligned} \angle QPS + \angle QRS &= 360^\circ - \frac{1}{2}(\angle A + \angle B + \angle C + \angle D) \\ &= 360^\circ - \frac{1}{2} \times 360^\circ \\ &= 360^\circ - 180^\circ \\ &= 180^\circ \end{aligned}$$

So, PQRS is a quadrilateral whose opposite angles are supplementary.

Hence, the correct option is (d).

Solution 15: (d) parallelogram

The figure formed by joining the mid points of the adjacent sides of a quadrilateral is a parallelogram.

Solution 16: (b) Square

The figure formed by joining the mid points of the adjacent sides of a square is a square.

Solution 17: (d) parallelogram.

The figure made by joining the mid points of the adjacent sides of a parallelogram is a parallelogram.

Solution 18: (a) rhombus

The figure formed by joining the mid points of the adjacent sides of a rectangle is a rhombus.

Solution 19: (c) Rectangle

The figure formed by joining the mid points of the adjacent sides of a rhombus is a rectangle.

Solution 20: (d) diagonals of ABCD are perpendicular to each other.

Solution 21: (c) diagonals of ABCD are equal

Since, ABCD is a rhombus. So, the sides AB, BC, CD and AD are equal.

Now, in ΔPQS , we have

$$DC = \frac{1}{2}QS \quad \dots (1) \text{ [Using mid - point theorem]}$$

Similarly, in ΔPSR ,

$$BC = \frac{1}{2}PR \quad \dots (2)$$

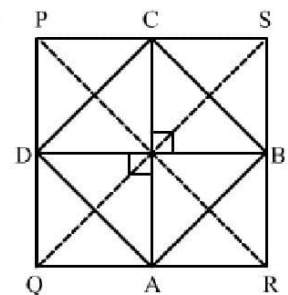
As, $BC = DC$

$$\Rightarrow \frac{1}{2}QS = \frac{1}{2}PR \quad \text{[From (1) and (2)]}$$

So, $QS = PR$

Thus, the diagonals of PQRS are equal.

Hence, the correct option is (c).



Solution 22: (d) diagonals of ABCD are equal and perpendicular

The quadrilateral formed by joining the mid-points of the sides of a quadrilateral with diagonals equal and perpendicular is square.

Hence, the correct option is (d).

Solution 23: (c) 72°

Let ABCD be a parallelogram.

$$\therefore \angle A = \angle C \text{ and } \angle B = \angle D \quad [\text{Opposite angles}]$$

$$\text{Let } \angle A = x \text{ and } \angle B = \frac{2}{3}x$$

$$\therefore \angle A + \angle B = 180^\circ \quad [\text{Adjacent angles are supplementary}]$$

$$\Rightarrow x + \frac{2}{3}x = 180^\circ$$

$$\Rightarrow \frac{5}{3}x = 180^\circ$$

$$\Rightarrow x = 108^\circ$$

$$\therefore \angle B = \frac{2}{3} \times 108^\circ = 72^\circ$$

Hence, $\angle A = \angle C = 108^\circ$ and $\angle B = \angle D = 72^\circ$

Solution 24: (c) 112°

Explanation:

Let ABCD is a parallelogram.

$$\therefore \angle A = \angle C \text{ and } \angle B = \angle D \quad [\text{Opposite angles}]$$

Let $\angle A$ be the smallest angle whose measure is x .

$$\therefore \angle B = (2x - 24)^\circ$$

$$\text{Now, } \angle A + \angle B = 180^\circ \quad [\text{Adjacent angles are supplementary}]$$

$$\Rightarrow x + 2x - 24^\circ = 180^\circ$$

$$\Rightarrow 3x = 204^\circ$$

$$\Rightarrow x = 68^\circ$$

$$\therefore \angle B = 2 \times 68^\circ - 24^\circ = 112^\circ$$

Hence, $\angle A = \angle C = 68^\circ$ and $\angle B = \angle D = 112^\circ$

Solution 25: (c) Trapezium

Let the angles be $3x$, $7x$, $6x$ and $4x$.

$$\text{Then } 3x + 7x + 6x + 4x = 360^\circ$$

$$\therefore x = 18^\circ$$

Thus, the angles are $3 \times 18 = 54^\circ$, $7 \times 18 = 126^\circ$, $6 \times 18 = 108^\circ$ and $4 \times 18 = 72^\circ$.

$$\text{But } 54 + 126 = 180 \text{ and } 72 + 108^\circ = 180^\circ$$

$\therefore$ ABCD is a trapezium.

Solution 26: (c) Opposite angles are bisected by the diagonals.

Solution 27: (c) Rectangle

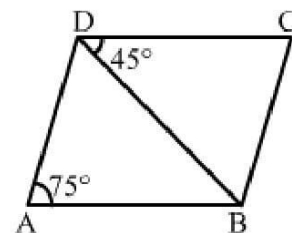
If APB and CQD are two parallel lines, then the bisectors of $\angle APQ$, $\angle BPQ$, $\angle CQP$ and $\angle PQD$ enclose a rectangle.

Solution 28: (c) 60°

$$\angle BAD = \angle BCD = 75^\circ \quad [\text{Opposite angles are equal}]$$

$$\text{In } \triangle BCD, \angle C = 75^\circ$$

$$\therefore \angle CBD = 180^\circ - (75^\circ + 45^\circ) = 60^\circ$$



Solution 29: (c) $A < B$

Let h be the height of parallelogram.

Then clearly, $h < b$

$$\therefore A = a \times h < a \times b = B$$

Hence, $A < B$

Solution 30: (b) $AF = 2 AB$

In parallelogram ABCD,

$AB \parallel DC$

$\angle DCE = \angle EBF$ [Alternate interior angles]

In $\triangle DCE$ and $\triangle BFE$, we have:

$\angle DCE = \angle EBF$ [Proved above]

$\angle DEC = \angle BEF$ [Vertically opposite angles]

$CE = BE$ [Given]

i.e., $\triangle DCE \cong \triangle BFE$ [By ASA congruence rule]

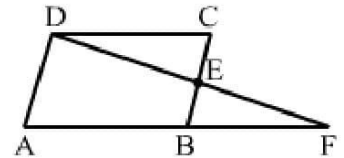
$\therefore DC = BF$ [CPCT]

But $DC = AB$, as ABCD is a parallelogram.

$\therefore DC = AB = BF$...(1)

Now, $AF = AB + BF$

$\Rightarrow AF = AB + AB = 2AB$ [From (1)]

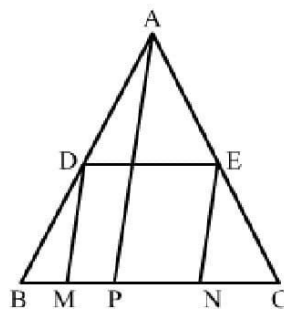


Solution 31: (b) a parallelogram

Given: In $\triangle ABC$, M, N, D and E are the mid-points of BP, CP, AB and AC, respectively.

In $\triangle ABP$,

$\therefore D$ and M are the mid-points of AB , and BP , respectively. [Given]



$\therefore DM = \frac{1}{2} AP$ and $DM \parallel AP$... (1) [Mid – point theorem]

Again, in $\triangle ACP$,

$\therefore$ E and N are the mid-points of AC, and CP, respectively. [Given]

$$\therefore EN = \frac{1}{2}AP \text{ and } EN \parallel AP \dots (2) \quad [\text{Mid - point theorem}]$$

From (1) and (2), we get

$$DM = EN \text{ and } DM \parallel EN$$

But this a pair of opposite sides of the quadrilateral DENM.

So, DENM is a parallelogram.

Hence, the correct option is (b).

Solution 32: (b) $\frac{1}{2}(a + b)$

Suppose ABCD is a trapezium.

Draw EF parallel to AB.

Join BD to cut EF at M.

Now, in $\triangle DAB$, E is the midpoint of AD and $EM \parallel AB$.

$\therefore$ M is the mid-point of BD and EM

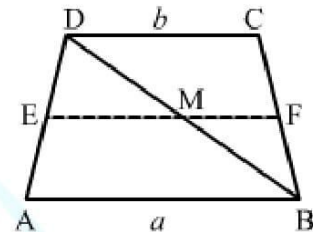
$$EM = \frac{1}{2} \times a$$

Similarly, M is the mid-point of BD and $MF \parallel DC$.

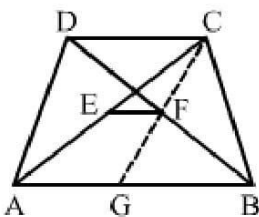
i.e., F is the midpoint of BC and MF

$$MF = \frac{1}{2} \times b$$

$$\therefore EF = EM + MF = \frac{1}{2}(a + b)$$



Solution 33: (d) $\frac{1}{2}(AB - CD)$



Join CF and produce it to cut AB at G.

Then $\triangle CDF \cong \triangle GBF$ [$\because DF = BF, \angle DCF = \angle BGF$ and $\angle CDF = \angle GBF$]

$\therefore CD = GB$

Thus, in $\triangle CAG$, the points E and F are the mid points of AC and CG, respectively.

$$\therefore EF = \frac{1}{2}(AG) = \frac{1}{2}(AB - GB) = \frac{1}{2}(AB - CD)$$

Solution 34: (c) 90°

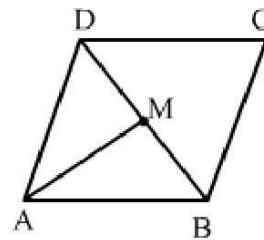
$$\angle B = \angle D$$

$$\Rightarrow \frac{1}{2}\angle B = \frac{1}{2}\angle D$$

$$\Rightarrow \angle ADB = \angle ABD$$

$\therefore \triangle ABD$ is an isosceles triangle and M is the midpoint of BD. We can also say that M is the median of $\triangle ABD$.

$\therefore AM \perp BD$ and, hence, $\angle AMB = 90^\circ$



Solution 35: (c) $AC^2 + BD^2 = 4AB^2$

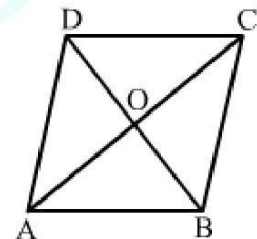
We know that the diagonals of a rhombus bisect each other at right angles.

$$\text{Here, } OA = \frac{1}{2}AC, OB = \frac{1}{2}BD \text{ and } \angle AOB = 90^\circ$$

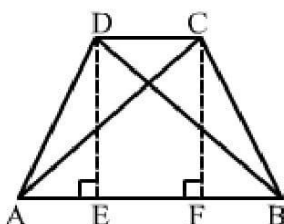
$$\text{Now, } AB^2 = OA^2 + OB^2$$

$$\Rightarrow AB^2 = \frac{1}{4}(AC)^2 + \frac{1}{4}(BD)^2$$

$$\therefore 4AB^2 = (AC^2 + BD^2)$$



Solution 36: (d) $BC^2 + AD^2 + 2AB \cdot CD$



Draw perpendicular from D and C on AB which meets AB at E and F, respectively.

$\therefore$ DEFC is a parallelogram and $EF = CD$.

In $\triangle ABC$, $\angle B$ is acute.

$$\therefore AC^2 = BC^2 + AB^2 - 2AB \cdot AE$$

In $\triangle ABD$, $\angle A$ is acute.

$$\therefore BD^2 = AD^2 + AB^2 - 2AB \cdot AF$$

$$\therefore AC^2 + BD^2 = (BC^2 + AD^2) + (AB^2 + AB^2) - 2AB(AE + BF)$$

$$= (BC^2 + AD^2) + 2AB(AB - AE - BF)$$

$$[\because AB = AE + EF + FB \text{ and } AB - AE = BE]$$

$$= (BC^2 + AD^2) + 2AB(BE - BF)$$

$$= (BC^2 + AD^2) + 2AB \cdot EF$$

$$\therefore AC^2 + BD^2 = (BC^2 + AD^2) + 2AB \cdot CD$$

Solution 37: (d) 1:1

Area of a parallelogram = base $\times$ height

If both parallelograms lie on the same base and between the same parallels, then their heights are the same.

So, their areas will also be the same.

Solution 38: (b) $\frac{1}{3} AC$

Let G be the mid-point of FC. Join DG.

In $\triangle BCF$, D is the mid-point of BC and G is the mid-point of FC.

$$\therefore DG \parallel BF$$

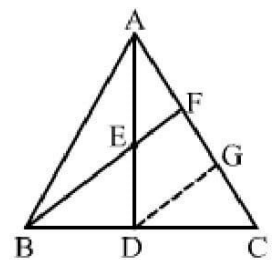
$$\Rightarrow DG \parallel EF$$

In $\triangle ADG$, E is the mid-point of AD and $EF \parallel DG$.

i.e., F is the mid-point of AG.

Now, $AF = FG = GC$ [$\because$ G is the mid-point of FC]

$$\therefore AF = \frac{1}{3} AC$$



Solution 39: (a) 40°

$$\angle OAD = \angle OCB = 30^\circ \quad [\text{Alternate interior angles}]$$

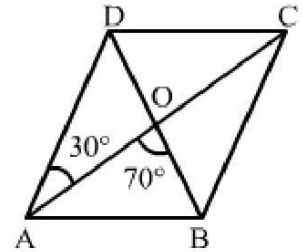
$$\angle AOB + \angle BOC = 180^\circ \quad [\text{Linear pair of angles}]$$

$$\therefore \angle BOC = 180^\circ - 70^\circ = 110^\circ \quad [\angle AOB = 70^\circ]$$

In $\triangle BOC$, we have:

$$\angle OBC = 180^\circ - (110^\circ + 30^\circ) = 40^\circ$$

$$\therefore \angle DBC = 40^\circ$$



Solution 40: (c) I and II

Solution 41: (b) II and III

Solution 42: Since, the opposite angles of the quadrilateral PQRS are equal.

$\therefore$ Quadrilateral PQRS is a parallelogram.

$$\Rightarrow SR = PQ \quad [\text{Opposite sides of a parallelogram are equal.}]$$

$$\therefore SR = PQ = 2 \text{ cm}$$

Solution 43: No, the statement is incorrect because the diagonals of a parallelogram bisect each other.

Solution 44: Since, in quadrilateral PQRS, $\angle P + \angle S = 180^\circ$.

i.e., the sum of adjacent angles is 180° .

So, PQRS is a parallelogram.

Solution 45: No, the statement is false because if all the four angles of a quadrilateral are less than 90° , then the sum of all four angles will be less than 360° .

Solution 46: Yes, the statement is true because all the angles of a quadrilateral such as rectangle and square are right angles.

Solution 47: No, the statement is false because if all angles are greater than 90° , then the sum of four obtuse angles will be greater than 360° .

Solution 48: Since, the sum of all angles (i.e. $70^\circ + 115^\circ + 60^\circ + 120^\circ = 365^\circ$).
So, we cannot form a quadrilateral with these angles.

Solution 49: We know that, sum of all angles in a quadrilateral is 360° .

Let each angle of the quadrilateral be x .

$$x + x + x + x = 360^\circ$$

$$\Rightarrow 4x = 360^\circ$$

$$\Rightarrow x = 90^\circ$$

$\Rightarrow$ All angles of the quadrilateral are 90° .

Hence, given quadrilateral is a rectangle.

Solution 50:

In $\triangle ABC$,

Since, D and E are respectively the mid-points of sides AB and BC. (Given)

So, $DE = \frac{1}{2}AC$ (Using mid-point theorem)

But $AC = 3.6$ cm (Given)

$$\text{or, } DE = \frac{1}{2} (3.6)$$

$$\text{or, } DE = 1.8 \text{ cm}$$

Hence, the length of DE is 1.8 cm.

Solution 51: Since, the diagonals PR and QS of quadrilateral PQRS bisect each other.

So, PQRS is a parallelogram.

$$\text{Now, } \angle Q + \angle R = 180^\circ \quad \angle Q + \angle R = 180^\circ$$

[Adjacent angles are supplementary]

$$\Rightarrow 56^\circ + \angle R = 180^\circ$$

$$\Rightarrow \angle R = 180^\circ - 56^\circ$$

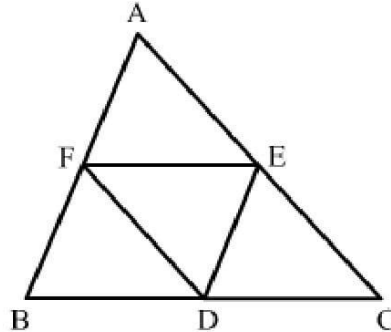
$$\Rightarrow \angle R = 124^\circ$$

$$\Rightarrow 56^\circ + \angle R = 180^\circ$$

$$\Rightarrow \angle R = 180^\circ - 56^\circ$$

$$\Rightarrow \angle R = 124^\circ$$

Solution 52:



Given: Parallelograms BDEF and AFDE.

Since, $BF = DE$...(1) [Opposite sides of parallelogram BDEF]

And, $AF = DE$...(2) [Opposite sides of parallelogram AFDE]

From (1) and (2), we get

$$AF = FB$$

Solution 53: (a) if the question can be answered by one of the given statements alone and not by the other;

We know that if the diagonals of a quadrilateral bisect each other, then it is a parallelogram.

$\therefore$ Statements I be the answer.

If the diagonals of a quadrilateral are equal, then it is not necessarily a parallelogram.

$\therefore$ Statements II does not give the answer.

Hence, the correct answer is (a).

Solution 54: (c) if the question can be answered by both the statements together but not by any one of the two;

Clearly, statements I alone is not sufficient to answer the given question.

Also, II alone is not sufficient to answer the given question.

However, both I and II together will give the answer.

∴ Hence, the correct answer is (c).

Solution 55: (c) if the question can be answered by both the statements together but not by any one of the two;

When the diagonals of a parallelogram are equal, it is either a rectangle or a square.

Also, if the diagonals intersect at a right angle, then it is a square.

∴ Both I and II together will give the answer.

Hence, the correct answer is (c).

Solution 56: (b) if the question can be answered by either statement alone;

We know that a quadrilateral is a parallelogram when either I or II holds true.

Hence, the correct answer is (b).

Solution 57: (a) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

Fourth angle = $360^\circ - (130^\circ + 70^\circ + 60^\circ) = 100^\circ$

Clearly, reason (R) and assertion (A) are both true and the reason gives the assertion.

Hence, the correct answer is (a).

Solution 58: (a) Both Assertion and Reason are true and Reason is a correct explanation of Assertion.

Clearly, reason (R) and assertion (A) are both true and reason (R) gives assertion (A).

Hence, the correct answer is (a).

Solution 59: (b) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

Clearly, reason (R) and assertion (A) are both true.

But reason (R) does not give assertion (A).

Hence, the correct answer is (b).

Solution 60: (d) Assertion is false and Reason is true.

We can easily prove reason (R). So, reason (R) is true.

Clearly, assertion (A) is false (as every parallelogram is not necessarily a rectangle).

Hence, the correct answer is (d).

Solution 61: (b) Both Assertion and Reason are true but Reason is not a correct explanation of Assertion.

Clearly, assertion (A) is true.

We can easily prove reason (R). So, (R) is also true.

But, reason (R) does not give assertion (A).

Hence, the correct answer is (b)

Solution 62:

(a) - (q),

(b) - (r),

(c) - (s),

(d) - (p)

Solution:

(a) - (r),

(b) - (s),

(c) - (p),

(d) - (q)

$$(a) PQ = \frac{1}{2}(AB + CD) = \frac{1}{2} \times 17 = 8.5 \text{ cm}$$

$$(b) OR = \frac{1}{2}PR = \frac{1}{2} \times 13 = 6.5 \text{ cm}$$